

AVISHKARANA INTERNSHIP PROGRAM

Mini Project Submission Format

Student Name: NARAYANA THARUN KUMAR

College Name: NRI INSTITUTE OF TECHNOLOGY

Branch & Year: Information Technology - 3rd Year

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Project Title: Student Performance Analytics Dashboard

Project Domain: Data Analysis / Python / Flask / ReactJS

Project Objective

Develop a Student Performance Analytics Dashboard that analyzes student marks, attendance, and grades. It provides interactive visualizations and insights to help understand student performance.

Components / Software Used

Python 3.x, Flask, ReactJS (Vite), Pandas, NumPy, Matplotlib, SQLite, VS Code, Git, GitHub, CSV Dataset.

Working Principle

The application loads student data from a CSV file, cleans and processes it, calculates total marks, averages and grades, stores records in SQLite, exposes REST APIs using Flask, and displays analytics through a React dashboard.

Results

The dashboard successfully displays KPIs, charts, grade distribution, top performers and student records.

Project Submission

Attached project screenshots, output screenshots, source code and documentation.

Source Code Link

<https://github.com/tarunnarayana18/miniproject.git>

Demo Video

Optional

Challenges Faced

Data cleaning, API integration, dashboard development, and frontend-backend communication.

Future Scope

Integrate machine learning for prediction, user authentication, larger datasets and cloud deployment.

Declaration

I declare that this project is my own work.

Student Signature: NARAYANA THARUN KUMAR

Date: 29 June 2026